

27.07.2026

PRACTICAL-2

UNDERSTANDING DIFFERENT TYPES OF NETWORK CABLE

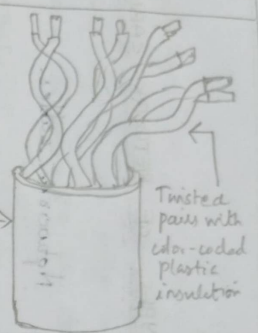
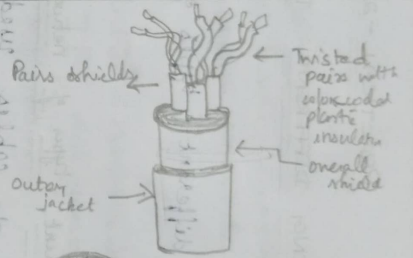
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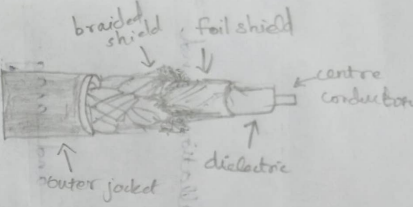
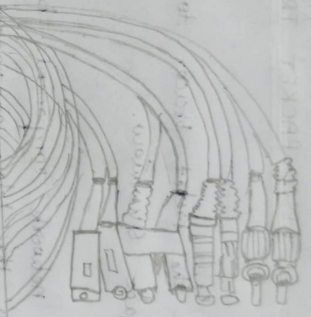
Study of different types of Network cables.

a) Understand different types of network cable.

Different types of cables used in networking are:

- 1) Unshielded Twisted Pair (UTP) cable.
- 2) Shielded Twisted Pair (STP) cable
- 3) Coaxial Cable
- 4) Fibre optic cable.

Cable type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use	Image
UTP	Category 3	10bps	→ Advantages: • Cheaper in cost • Easy to install as they have a smaller overall diameter. → Disadvantages:- • More prone to (EMI) Electromagnetic interference and noise.	10-Base-T Ethernet	 Twisted pairs with color-coded plastic insulation Outer jacket
	Category 5	upto 100 Mbps		Fast Ethernet, Gigabit Ethernet	
	Category 5e	1 Gbps		Fast Ethernet, Gigabit Ethernet	
STP	Category 6, 6a	10 Gbps	→ Advantages: • Shielded • Faster than UTP • Less susceptible to noise and interference.	Gigabit Ethernet, 10G Ethernet (SSM) widely used in data centres.	 Pairs shielded Twisted pairs with color-coded plastic insulation Overall shield Outer jacket
SSTP	Category 7	10 Gbps		Gigabit Ethernet, 10G Ethernet (loom)	

Coaxial cable	RG-6 RG-59 RG-11	10-100Mbps	→ Advantages • High bandwidth • Immune to interference • Low loss bandwidth • Versatile → Disadvantages • Limited distance • Cost • High is bulky	Speed of signal is 500m Television network High speed internet connections	
Fibre optics cable	Single mode Multi mode	100Gbps	→ Advantages • High speed • High bandwidth • High security • Long distance → Disadvantages • Expensive • Requires skilled installers	Maximum distance of fibre optics cable is around 100 meters	

RESULT:

Thus we have studied the different types of network cables.

Signature